PH102: Physics-2

Electrodynamics

Lecture # 11

Electrostatics – Electric fields in matter – Electric displacement and linear dielectrics

14.02.2024

D. Pamu and A. K. Sharma

Previous lecture

- Polarization results in the dipole moment, $\mathbf{p} = \alpha\mathbf{E}.$
- Torque on a permanent dipole in uniform field is $\mathbf{N} = \mathbf{p} \times \mathbf{E}$
- Force on a permanent dipole in a non-uniform field is $\mathbf{F} = q\Delta\mathbf{E} = (\mathbf{p} \cdot \boldsymbol{\nabla})\mathbf{E}$
- Potential at points far from the dipole is $V(\mathbf{r}) = \dfrac{1}{4\pi\varepsilon_0}\dfrac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$
- The electric field due to a dipole is $\mathbf{E}_{dip}(\mathbf{r}) = \dfrac{1}{4\pi\varepsilon_0 r^3}[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$
- The bound surface charge density is $\sigma_b = \mathbf{P}\cdot\hat{n}$
- The bound volume charge density $\rho_b = -\boldsymbol{\nabla}\cdot\mathbf{P}$

The field inside a dielectric

What kind of dipole are we actually dealing with, "pure" dipole or "physical" dipole?

- In developing the theory of bound charges, we assumed the pure dipole.
- An actual polarized dielectric consists of physical dipoles, extremely tiny ones.
- Discrete molecular dipoles are represented by a continuous density function **P**.
- Outside the dielectric, there is no real problem (far away from the molecules).
- Inside the dielectric, however, all the dipoles are very close together.

E inside the matter is very complicated at the ***microscopic level***,

- near an electron, the field is gigantic, a short distance away it may be small or may point in a totally different direction.
- as the atoms move about, the field will change entirely.

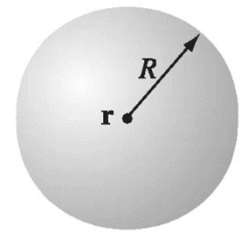
We can ignore the microscopic bumps and wrinkles in the electric field inside the matter, and concentrate on the macroscopic field.

(Contd…)

The ***macroscopic field*** **-** the **average field** over regions of a system large enough to **contain many thousand of atoms** and yet **small enough** compared to the dimension of the system.

Calculate the macroscopic field at some point *r* within a dielectric.

Average the true (microscopic) field over an appropriate volume, say, a thousand times the size of a molecule.



$$\mathbf{E} = \mathbf{E}_{out} + \mathbf{E}_{in}.$$

the average field over the sphere due to all charges outside.

the average due to all charges inside.

The average field over a sphere, produced by charges outside is equal to the field they produce at the center of the sphere.

The dipoles inside the sphere are too close and therefore, treating the continuous distribution of dipoles

$$\mathbf{E}_{in} = \frac{\rho r}{3\varepsilon_0}\hat{\mathbf{r}} = -\frac{1}{4\pi\varepsilon_o}\frac{\mathrm{p}}{R^3}\hat{\mathbf{r}} = -\frac{1}{3\varepsilon_o}\mathbf{P}$$

(Contd…)

$\mathbf{E}_{out}$ is the field at *r* due to the dipoles exterior to the sphere. These are far enough away, we can write the corresponding potential,

$$V_{out} = \frac{1}{4\pi\varepsilon_o} \int_{outside} \frac{\hat{\mathbf{r}} \cdot \mathrm{P}(\mathbf{r}')}{r^2} d\tau'$$

So, the contributions from the above two equations give the total electric field.

However, assuming the sphere is small enough (**P** does not vary significantly over its volume), the term *left out* of the integral is the field at the center of a *uniformly polarized sphere* if the integral runs over the *entire* volume of the dielectric (solved example 3, lecture 10) i.e.,

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_o} \int \frac{\hat{\mathbf{r}} \cdot \mathrm{P}(\mathbf{r}')}{r^2} d\tau'$$

But this is what we used to calculate the field inside the dielectric !!!

No matter how crazy the actual microscopic charge configuration, we can replace it by a nice smooth distribution of perfect dipoles, if all we want is the macroscopic (average) field.

Free charge

The effect of polarization is to produce bound charges,

- $\rho_b = -\nabla \cdot \mathbf{P}$ within the dielectric and
- $\sigma_b = \mathbf{P} \cdot \hat{n}$ on the surface.

The field due to polarization of the medium is just the field of these bound charges. Now, treat the field caused by both bound charges and free charge within the dielectric.

The total charge density can be written as

$$\rho = \rho_b + \rho_f$$

What is a free charge?

- The field due to everything *else* (any charge that is *not* a result of polarization).
- The free charge might consist of electrons on a conductor or ions embedded in the dielectric material etc.

The electric displacement

Let us apply gauss's law in the presence of dielectrics to the total charge density,

$$\varepsilon_o(\nabla \cdot \mathbf{E}) = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f$$

E is the total field, not just the portion generated by polarization. Rearranging,

$$\nabla \cdot (\varepsilon_o \mathbf{E} + \mathbf{P}) = \rho_f$$

We define,

$$\mathbf{D} \equiv \varepsilon_o \mathbf{E} + \mathbf{P}$$

where **D** is known as electric displacement.

The Gauss's law in terms of **D** can be written as

$$\nabla \cdot \mathbf{D} = \rho_f$$

and in integral form,

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{fenc}$$

Q_{fenc} is the total free charge enclosed in the volume.

These relations are referred to as constitutive relations in the literature.

(Contd…)

Important question: While deriving the relation $\nabla \cdot \mathbf{D} = \rho_f$, what happened to the surface charge?

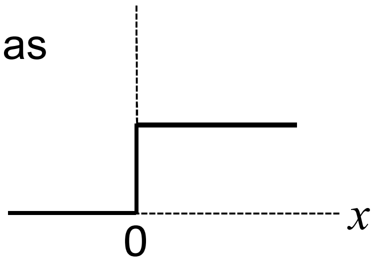
- We cannot apply Gauss's law precisely at the surface of a dielectric, for here ρ_b blows up.
- Picture the edge of the dielectric as having some finite thickness, within which the polarization tapers off to zero (probably a more realistic model).
- Then there is no surface bound charge, ρ_b varies rapidly but smoothly within this "skin," and Gauss's law can be safely applied everywhere.
- The integral form anyway is free from this "defect."

The polarization drops abruptly to zero outside the material, so its derivative is a delta function. The surface bound charge is precisely this term - in this sense it is actually included in ρ_b, but we ordinarily prefer to handle it separately as σ_b.

Step function is defined as

$$\theta(x) = \begin{cases} 1, x > 0 \\ 0, x \le 0 \end{cases}$$

$$\frac{d\theta}{dx} = \delta(x)$$



(Contd…)

- Gauss's law, in the context of dielectrics, *makes reference only to free charges,* and free charge is the stuff we control.
- When we put the free charge in place, a certain polarization automatically ensues, and this polarization produces the bound charge.
- In a typical problem, therefore, we know ρ_f, but we do not (initially) know ρ_b.
- In particular, whenever the requisite symmetry is present, we can immediately calculate **D** by the standard Gauss's law methods.

A deceptive parallel between D and E

The equation $\nabla \cdot \mathbf{D} = \rho_f$

- looks just like Gauss's law,
- only the total charge density ρ is replaced by the free charge density ρ_f, and
- **D** is substituted for ε_o**E**.

Do not conclude that **D** is "just like" **E.**

"To solve problems involving dielectrics, you just forget all about the bound charge - calculate the field as you ordinarily would, only call the answer **D** instead of **E**."

There is no "Coulomb's law" for **D**,

$$\mathbf{D}(\mathbf{r}) \neq \frac{1}{4\pi} \int \frac{\hat{\mathbf{r}}}{r^2} \rho_f(\mathrm{r}')\, d\tau'$$

The divergence alone is insufficient to determine a vector field; you need to know the curl as well (the **Helmholtz theorem** guarantees that the field is uniquely determined by its divergence and curl).

(Contd…)

In the case of electrostatic fields, the curl of **E** is always zero. But the curl of **D** is not always zero.

$$\mathbf{\nabla} \times \mathbf{D} = \varepsilon_0(\mathbf{\nabla} \times \mathbf{E}) + (\mathbf{\nabla} \times \mathbf{P}) = (\mathbf{\nabla} \times \mathbf{P})$$

The curl of P need not vanish.

Since **the curl of D is not always zero, D cannot be** expressed as the gradient of a scalar.

Advice : If the problem exhibits *spherical, cylindrical, or plane symmetry,* then you can get **D directly from the generalized Gauss's law**. If the requisite symmetry is absent, you have to think another approach, you must not assume D is determined exclusively by free charge.

Boundary Conditions

The electric field is not continuous at a surface with charge density σ. What about **D**?

Use the approach similar to the one used to obtain the boundary conditions for the electric field by considering it just above and below the surface of interest.

Gauss's law for the electric displacement states that

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = Q_{fenc} = \sigma_f A$$

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = \int \mathbf{D}^{\perp}_{above} \cdot \hat{n} da + \int \mathbf{D}^{\perp}_{below} \cdot (-\hat{n}) da = D^{\perp}_{above} A - D^{\perp}_{below} A$$

Therefore, $\boxed{D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f}$

In the absence of free charge, D$^{\perp}$ is continuous.

The normal component of D is discontinuous by an amount σ_f at any boundary.

$D^{\perp}$ denotes the component of **D** that is perpendicular to the surface immediately above or below.

(Contd…)

Using, $\nabla \times \mathbf{D} = \varepsilon_0(\nabla \times \mathbf{E}) + (\nabla \times \mathbf{P}) = (\nabla \times \mathbf{P})$,

and applying Stokes' theorem, we get

$$\oint \mathbf{D} \cdot d\mathbf{l} = \oint \mathbf{P} \cdot d\mathbf{l}$$

$$\oint_l \mathbf{D} \cdot d\mathbf{l} = \int \mathbf{D}^{\parallel}_{above} \cdot \hat{l} dl + \int \mathbf{D}^{\parallel}_{below} \cdot (-\hat{l}) dl = D^{\parallel}_{above} l - D^{\parallel}_{below} l$$

Similarly,

$$\oint_l \mathbf{P} \cdot d\mathbf{l} = \int \mathbf{P}^{\parallel}_{above} \cdot \hat{l} dl + \int \mathbf{P}^{\parallel}_{below} \cdot (-\hat{l}) dl = P^{\parallel}_{above} l - P^{\parallel}_{below} l$$

Therefore, $\boxed{D^{\parallel}_{above} - D^{\parallel}_{below} = P^{\parallel}_{above} - P^{\parallel}_{below}}$

In the absence of polarization, $\mathbf{D}^{\parallel}$ is continuous.

$D^{\parallel}$ denotes the component of **D** that is parallel to the surface immediately above or below.

The parallel component of D is discontinuous at any boundary.

Linear dielectrics

For many substances, the polarization is proportional to the field, provided **E** is not too strong,

$$\mathbf{P} = \varepsilon_o \chi_e \mathbf{E}$$

χ_e is the electric susceptibility of the material.

E is the total field (in part to free charges and in part to the polarization itself).

It is the response of the material to the external electric field that results in the linear or non-linear behavior of the material.

Therefore, in a linear dielectric medium,

$$\mathbf{D} = \varepsilon_o \mathbf{E} + \mathbf{P} = \varepsilon_o \mathbf{E} + \varepsilon_o \chi_e \mathbf{E} = \varepsilon_o (1 + \chi_e) \mathbf{E}$$

$$\mathbf{D} = \varepsilon \mathbf{E}$$

where,

$$\varepsilon = \varepsilon_o (1 + \chi_e)$$

ε is the permittivity of the material.

As a dimensionless quantity, we define

$$\epsilon_r = \frac{\varepsilon}{\varepsilon_o} = 1 + \chi_e$$

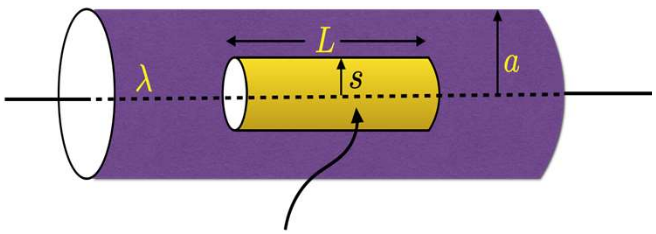
Dielectric constants for some substances

Material	Dielectric Constant	Material	Dielectric Constant
Vacuum	1	Benzene	2.28
Helium	1.000065	Diamond	5.7-5.9
Neon	1.00013	Salt	5.9
Hydrogen (H_2)	1.000254	Silicon	11.7
Argon	1.000517	Methanol	33.0
Air (dry)	1.000536	Water	80.1
Nitrogen (N_2)	1.000548	Ice (-30° C)	104
Water vapor (100° C)	1.00589	$KTaNbO_3$ (0° C)	34,000

where ϵ_r is called the relative permittivity, or dielectric constant, of the material.

Solved example 1

A long straight wire, carrying uniform line charge λ, is surrounded by rubber insulation out to a radius *a*. Find the electric displacement.



Applying, $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{fenc}$

(λL is the free charge).

we get,

$$D(2\pi s L) = \lambda L$$

$$\mathbf{D} = \frac{\lambda}{2\pi s}\hat{\mathbf{s}}$$

This expression holds both within the insulation and outside it **(Why?)**.

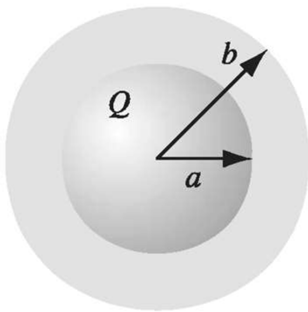
In the later region, **P** = 0 **(why?)**, so,

$$\mathbf{E} = \frac{1}{\varepsilon_o}\mathbf{D} = \frac{\lambda}{2\pi\varepsilon_o s}\hat{\mathbf{s}}$$

Inside the rubber the electric field can not be determined, since we do not know **P**.

Solved example 2

A metal sphere of radius *a* carries a charge *Q*. It is surrounded, out to radius *b*, by linear dielectric material of permittivity ε. Find the potential at the center (relative to infinity).



- To compute *V*, we need to know **E**.
- To find **E**, we might first try to locate the bound charge.
- We could get the bound charge from **P**.
- But we can't calculate **P** unless we already know **E** ($\mathbf{P} = \varepsilon_o \chi_e \mathbf{E}$).

The free charge *Q* is known, and the arrangement is spherically symmetric.

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{fenc} \Rightarrow \mathbf{D} = \frac{Q}{4\pi r^2}\hat{\mathbf{r}}$$

Inside the metal sphere, **E** = **P** = **D** = 0,

$$\mathbf{E} = \begin{cases} \dfrac{Q}{4\pi\varepsilon r^2}\hat{\mathbf{r}} \quad for \quad a < r < b \\ \dfrac{Q}{4\pi\varepsilon_o r^2}\hat{\mathbf{r}} \quad for \quad r > b \end{cases}$$

(Contd…)

The potential at the center is

$$V = -\int_{\infty}^{o} \mathbf{E}\cdot d\mathbf{l} = -\int_{\infty}^{b}\left(\frac{Q}{4\pi\varepsilon_o r^2}\right)dr - \int_{b}^{a}\left(\frac{Q}{4\pi\varepsilon\ \ r^2}\right)dr - \int_{a}^{0}(0)\,dr$$

$$V = \frac{Q}{4\pi}\left(\frac{1}{\varepsilon_o b} + \frac{1}{\varepsilon a} - \frac{1}{\varepsilon b}\right)$$

What is the polarization in the dielectric?

$$\mathbf{P} = \varepsilon_o\chi_e\mathbf{E} = \frac{\varepsilon_o\chi_e Q}{4\pi\varepsilon r^2}\hat{\mathbf{r}}.$$

And bound charges?

$$\rho_b = -\boldsymbol{\nabla}\cdot\mathbf{P} = 0.$$

$$\sigma_b = \mathrm{P}\cdot\hat{n} = \begin{cases} \dfrac{\varepsilon_o\chi_e Q}{4\pi\varepsilon b^2} & at\quad the\quad outer\quad surface \\ \dfrac{-\varepsilon_o\chi_e Q}{4\pi\varepsilon a^2} & at\quad the\quad inner\quad surface \end{cases}$$

The surface bound charge at *a* is negative ($\hat{n}$ points outward with respect to the dielectric, which is + $\hat{\mathbf{r}}$ at *b* but - $\hat{\mathbf{r}}$ at *a*). This is natural, since the charge on the metal sphere attracts its opposite in all the dielectric molecules. It is this layer of negative charge that reduces the field, within the dielectric, from $\frac{1}{4\pi\varepsilon_o}\left(\frac{Q}{r^2}\right)\hat{\mathbf{r}}$ to $\frac{1}{4\pi\varepsilon}\left(\frac{Q}{r^2}\right)\hat{\mathbf{r}}$

A dielectric is like an imperfect conductor: on a conducting shell the induced surface charge would be such as to cancel the field of Q *completely* in the region *a* < *r* < *b*: the cancellation is only partial.

Questions?